

CAMERA TECHNOLOGY

The modern-day smartphone is not just a simple communication device. Over the last two decades, phone makers have slowly expanded the smartphone's capabilities to the point that in 2021 it can not only make calls, act as a portable computer or a portable gaming device, but also attempt to double up as a pocket sized replacement for a bulky DSLR.

While smartphone cameras aren't there just yet, it will be disingenuous to not mention that they are definitely there or thereabouts. Years of incremental upgrades and perfection of camera technology has managed to ensure that smartphones of today encompass many features that were once only present in standalone cameras. In fact, in some cases, smartphone cameras even appear to have left DSLRs behind.

But the future of smartphone photography could well see camera phones comprehensively beat DSLRs when it comes to hardware and eventually performance. This is considering that DSLRs don't drastically improve their sensors. While OEMs like Samsung already have mobile camera lenses with huge 108MP sensors, in the future, we could easily see phones with even bigger sensors.

Samsung has already announced that it is working on developing a 600MP lens. This yet-to-be-named sensor will not only offer significantly improved still photography, but also improved 4K and 8K video recording. Further, its crazy megapixel count will likely enable it to facilitate extremely high, and never-before-seen optical zoom levels on smartphones.

But, as the Pixels and the iPhones have taught us over the years, photography on smartphones is not just about hardware. In fact, it's more about software than hardware. So, in the years to come, we'll see smartphones truly beating the DSLRs at their own game with the help of complex computational photography algorithms. These codes will be much more advanced than the ones that exist today, and as such, working in tandem with powerful hardware, will

be able to achieve far more impressive results than what is being achieved by current generation camera phones.

BATTERY LIFE

For the smartphone of today, the biggest bottleneck that exists is definitely the battery life. Despite being capable of handling complex tasks and acting as a standalone camera, or a portable gaming or even media consumption device, where the humble smartphone falls flat on its face is in the battery department. While there are a number of mainstream phones which offer huge batteries, they still don't appear to be good enough to handle all that the smartphone can do. But this could soon be a thing of the past.

Current generation smartphones use Lithium-Ion batteries which were first developed for commercial use in 1991. However, in 2021 a number of companies are working towards bringing wacky alternatives such as Aluminium-Air batteries. These batteries could provide weeks worth of use in a cycle, and can theoretically be charged in a matter of minutes.



Photo courtesy: Xiaomi

Charge to capacity in minutes

While this idea may seem a little far-fetched, it's actually quite close to reality. However, if you're looking for something much closer to reality, you can sample Graphene batteries. This as a material appears most likely to replace lithium-ion batteries in the next few years. These batteries could provide up to 60 per cent more capacity, and support much faster charging technologies than lithium-ion batteries. And all this, while being safer than current generation phone batteries.

WHAT WILL SMARTPHONES LOOK LIKE IN 2041?

Nothing like what we can imagine.

While forecasting the future of smartphones may be tricky, we believe when Digit celebrates its 40th anniversary, the smartphone of 2041 will hopefully be nothing like the smartphone of today. They will likely be science fiction gadgets that will not look, or work anything like the smartphones we use today.

Phone batteries will shrink, Samsung, Sony and LG will possibly find a way to make holographic displays that look as good as the real thing, and physical or even virtual keyboards and buttons will likely be replaced by gesture commands. It is entirely possible for smartphone makers to come up with Neuralink-like solutions that will be used as a futuristic way to control our phones with our brains. Meanwhile, the phone itself may shrink to the size of a tiny business card.

While there have been enough suggestions that the phone of the future may turn into a wearable, we believe that won't be the case. And even if it ends up being the case, that's a scenario we're certainly not rooting for at Digit.

As for what the smartphone of 2041 will be able to do? I'd say the possibilities are endless. Smartphones in twenty years will be more central to our lives than they are right now. They will replace our wallets and our house keys. They already have in a lot of countries owing to the pandemic. They will be more self-aware and be part of our lives in ways that we've only begun to imagine right now. From controlling the temperature in our rooms, to ordering food for us at the drop of a thought, the smartphone of 2041 will be more central to our daily way of life than ever before.

As mentioned above, it is truly very difficult to forecast just how exactly the phone of the future will look like, or what exactly it will do. But what we're sure of is that it will be a very familiar, yet very strange beast than the one we carry in our pockets today. 